

MYP 1 Design

PictoBlox Block Categories

Learning Objective

By the end of this lesson, you will be able to:

- Identify the different PictoBlox block categories.
- Tell what each category is used for.
- Create simple programs using different blocks.

What are Block Categories?

- PictoBlox groups similar coding blocks into different categories.
- Each category has a special job.
- Think of them like tools in a toolbox—each tool is used for a different task.

1. Motion Blocks: Motion blocks help a sprite move.

Examples

- Move steps
- Turn right
- Turn left
- Go to x: y:
- Glide

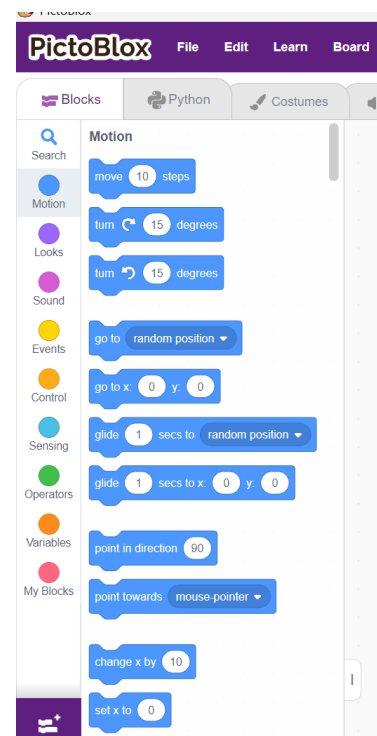
Activity

Try these blocks:

- When Green Flag Clicked
- Move 100 Steps

Challenge: Can you make the sprite:

- ✓ Move forward
- ✓ Turn right
- ✓ Move again

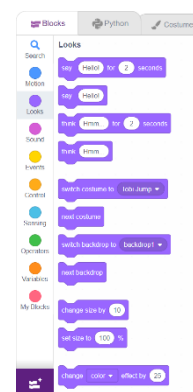


2. Looks Blocks: Looks blocks change how a sprite appears.

They can:

- Speak
- Think
- Change costumes
- Change size
- Show or hide

Examples



- Say "Hello!"
- Think "Hmm..."
- Change costume
- Hide
- Show

Activity: Create this program.

When Green Flag Clicked



Say

Hello!

for 2 seconds



Think

I like coding!

for 2 seconds

Challenge: Can your sprite say: "My name is ____."

3. Sound Blocks: Sound blocks allow the sprite to make sounds or play music.

Examples

- Play Sound
- Start Sound
- Stop All Sounds

Activity

Choose any sound.

When Green Flag Clicked

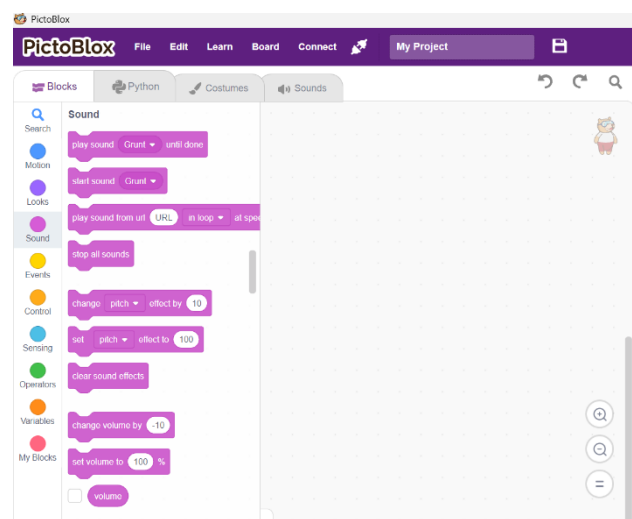


Play Sound

Challenge

Try a different sound.

Which one do you like most?



4. Events Blocks: Events tell the program when to start. They begin actions.

Examples

- When Green Flag Clicked
- When Sprite Clicked
- When Key Pressed

Activity

Use:
When Sprite Clicked



Say
"You clicked me!"

Try It

Click the sprite.

What happened?

Challenge

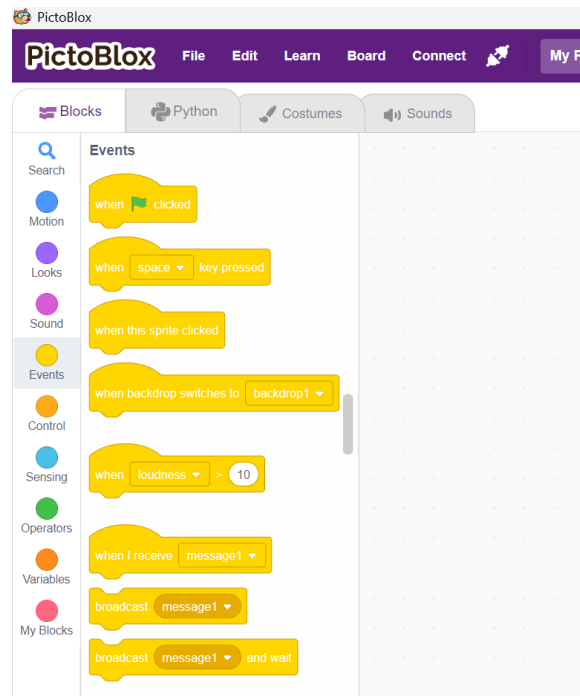
Try changing the event.

Instead of clicking the sprite,

use:

When Space Key Pressed

What happens?



5. Control Blocks: Control blocks manage the flow of the program.

They help repeat actions or make the sprite wait.

Examples

- Wait
- Repeat
- Forever

Activity

Create this:

When Green Flag Clicked

Repeat (5)

Move 20 Steps

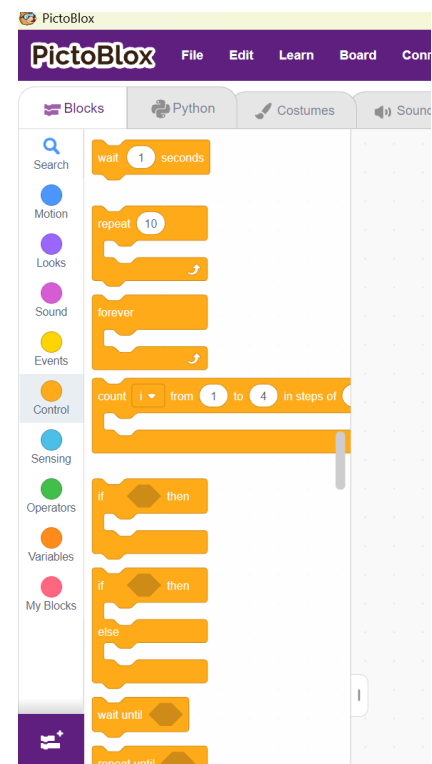
Wait 1 Second

Add:

Wait 1 Second

inside the Repeat block.

What changed?



6. Sensing Blocks: Sensing blocks help the sprite detect actions.

The sprite can know if:

- A key is pressed
- The mouse is touching it
- It is touching another object

Examples

- Touching Mouse Pointer?
- Key Pressed?
- Ask and Wait

Activity

Use:

When Green Flag Clicked

↓

Ask

"What is your name?"

↓

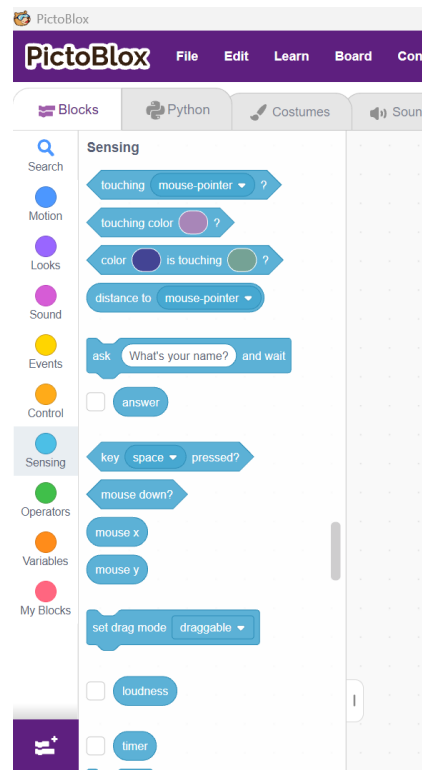
Wait

Challenge

Type your name.

Did the sprite wait until you answered?

Yes / No



7. Operators Blocks: Operators help perform calculations and compare values.

They can:

- Add
- Subtract
- Multiply
- Compare numbers
- Join words

Examples

- $5 + 3$
- $10 > 6$
- Join Hello + World
- Join "Hello" and "Friends!"

Activity

Use:

Join

Hello

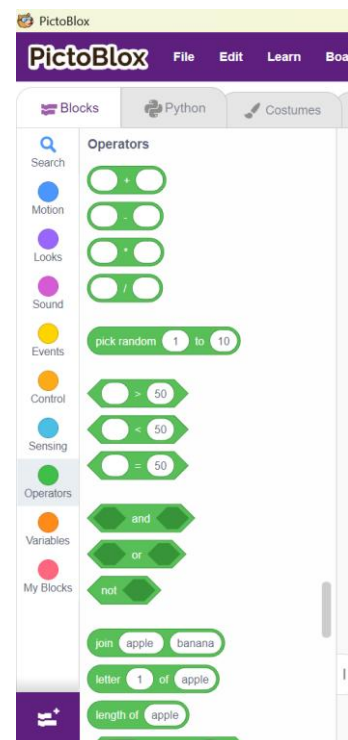
and

Friends!

inside a Say block.

Output

Hello Friends!



Challenge

Change the words.

Write your own message.

Quick Summary

Block Category	What it Does
Motion	Moves the sprite
Looks	Changes the sprite's appearance
Sound	Plays sounds
Events	Starts the program
Control	Repeats actions and waits
Sensing	Detects actions or asks questions
Operators	Performs calculations and joins text

Home Work Task:

Create your own program using **at least four different block categories**.

Your program should:

- Start with the Green Flag.
- Make the sprite say a message.
- Move the sprite.
- Wait for 1 second.
- Play a sound.

Reflection

1. Which block categories did you use?
2. Which category was your favourite?
3. What new thing did you learn today?
4. Which block category would you like to explore more?